

Capstone Progress

Simulation system

edge.py

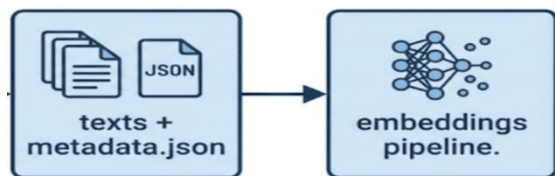
- Load image and pre-trained model
- -> image preprocessing (resize, split, and tokenizer)
- -> vision encoder
- -> projected visual embeddings
- -> send embeddings to cloud (HTTP API + json)

cloud.py

- receive projected visual embeddings
- -> rebuild tensor
- -> tokenize text prompt
- -> convert prompt tokens to text embeddings
- -> concatenate visual embeddings + text embeddings
- -> Generate ids
 - language decoder
 - -> lm head
- -> generated text answer

Cloud Service

Reconstruct embedding



Construct prompt



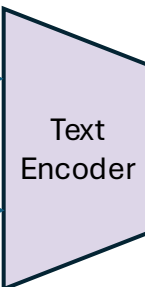
Generate text



Evaluation scripts

Output text

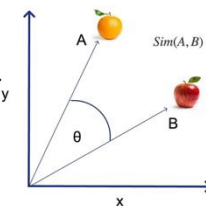
NanoVLM text



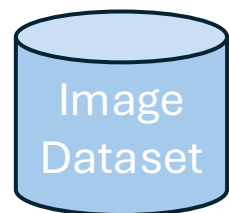
T1 T2 ... T_N

T1 T2 ... T_N

Cosine Similarity



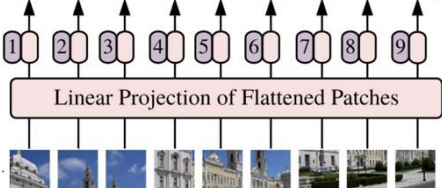
Edge layer



Resize + Patch



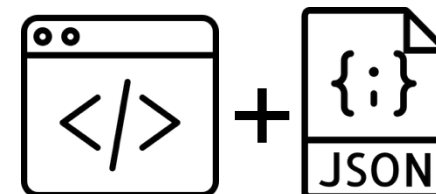
Split + Position Embedding



Encrypted data

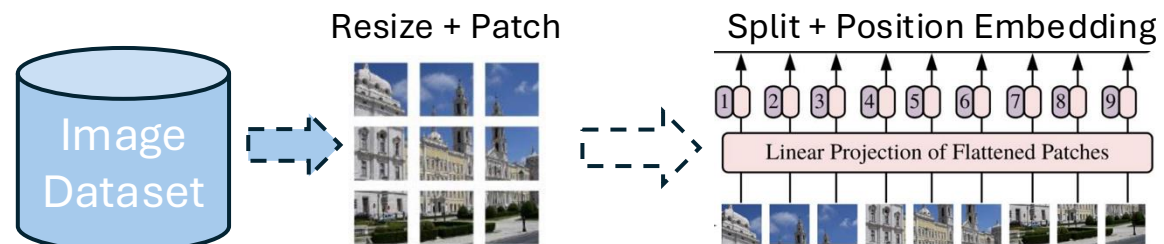


Metadata + encoder

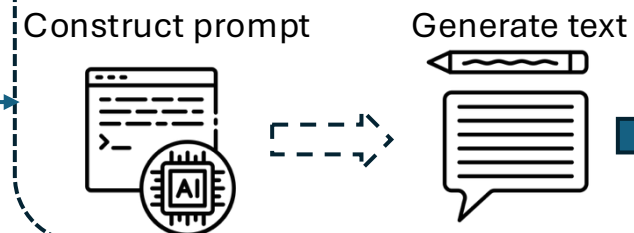


Edge layer

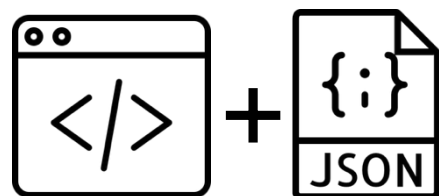
Visual Embedding Extraction



Original nanoVLM Inference



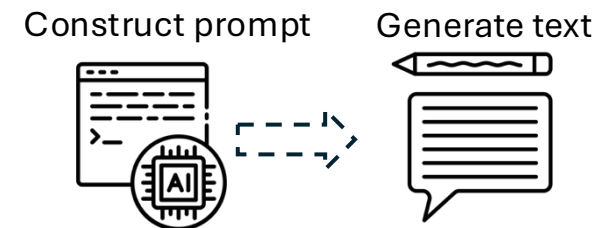
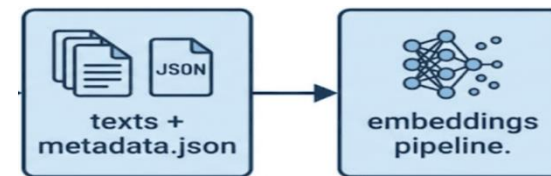
Serialized Embeddings + Metadata



Network(Lo)

Cloud Service

Reconstruct embedding



Output Comparison

Literature Review

- Current Problem and constrain -- Survey
- Privacy Preserving and Security
 - - Fine-tuning Phase – Encryption or FHE
 - - Inference Phase – split learning + Encryption + MAC
 - Authentication:
- Security Machanism:
 - Encryption:
 - Authentication:
 - Protocol:

Current Industry Practice 1

Direct raw image transmission via REST/RPC API

Client / camera / in-vehicle system → JPEG/PNG image → Base64 JSON or binary HTTPS POST → cloud inference service → text output

Main idea:

- The original image is directly uploaded to the cloud service.
- This is the common design used by many cloud vision APIs.
- The cloud performs image understanding, OCR, object detection, or captioning.

Payload format:

Base64 image inside JSON request, Multipart/form-data binary upload, application/octet-stream binary upload

Encryption: HTTPS / TLS protects data in transit

Authentication: API key, Bearer token / OAuth token, Service account / IAM credentials

Example:

Google Cloud Vision API

Azure Computer Vision

AWS Rekognition

Current Industry Practice 2

- Indirect cloud-storage transmission (offline / asynchronous)
- Client / edge device -> JPEG/PNG image -> HTTPS upload -> cloud storage bucket -> cloud function / inference service reads image -> model inference → text
- Main idea:
 - The original image is uploaded to object storage first, Then, the inference service later reads the image from storage. This is useful for batch processing, asynchronous inference, and large image files.
 - Encryption: In transit: HTTPS / TLS, At rest: server-side encryption (SSE-S3, SSE-KMS, or Azure/GCP managed encryption)
 - Authentication: Pre-signed URL (IAM role / service account), OAuth token, Access key / SAS token, depending on the cloud provider
- For instance:
 - AWS S3 + Lambda + Rekognition / custom model
 - Google Cloud Storage + Cloud Functions + Vision API
 - Azure Blob Storage + Azure Functions + Computer Vision

Current Industry Practice 3

- MQTT / IoT message broker transmission
- Camera / IoT device → JPEG frame → MQTT binary payload or Base64 payload → broker → cloud function / inference service → text
- Main idea:
 - Encryption : MQTT does **not normally encrypt messages by itself**. In real systems, MQTT is secured by running it over **TLS/SSL** (hardware support for AES is not commonly for embedded, ChaCha20 is a faster option)
 - Authentication: MQTT CONNECT packets can include **User Name** and **Password** fields. The broker can use these fields to authenticate the client. The MQTT 5.0 specification says implementations can use these fields directly, connect them to external authentication systems such as LDAP or OAuth, or use operating-system authentication.
- Source: <https://docs.oasis-open.org/mqtt/mqtt/v5.0/cos02/mqtt-v5.0-cos02>
- Used in : in-vehicle system

Literature review (basic idea)

- Image transmission protection methods are relevant background.
- Embedding transmission reduces raw-image exposure, but embeddings are still sensitive. (background)
- Narrow down to embedding encryption and authentication combinations (find research gap)

Image transmission protection methods are relevant background

- **IOT Device Encryption and Authentication:**
 - Embedded Systems Authentication and Encryption Using Strong PUF Modeling - Introducing that embedded devices face several security and performance limitations. (embedded encryption)
 - ChaCha20-Poly1305 authenticated encryption for high-speed embedded IoT applications (related to MQTT protocol)
 - Lightweight Internet of Things Device Authentication, Encryption, and Key Distribution Using End-to-End Neural Cryptosystems
- **Image Encryption:**
 - A Lightweight Image Encryption Algorithm Based on Secure Key Generation (based on AES)
- **Transmission protocol:** MQTT (classic protocol), what else?

Embedding transmission reduces raw-image exposure, but embeddings are still sensitive. (background)

- **Understanding deep image representations by inverting them** -- Inverts image representations to reconstruct visual content. This shown that, under white-box or partially known model settings, visual representations can be inverted to recover information about the original image.
- **Deep-BMI: Exploring Model Inversion Attacks in the Black-box Setting** -- Black-box model inversion can reconstruct semantically meaningful images without accessing target model parameters, although it usually requires repeated queries to the target model.
- **Deep Learning Model Inversion Attacks and Defenses: A Comprehensive Survey** -- how attackers can reconstruct private training data or sensitive information from deep learning models, and summarises existing defense methods.

Embedding encryption and authentication combinations

- **A privacy preserving split learning framework with adaptive inference selection for IoT security** -- Split Learning, Gaussian noise-based DP, pruning, quantization, TensorFlow Lite, and adaptive model selection.
- **A privacy-preserving multi-user retrieval system for multimodal artificial intelligence**
 - This paper shows that embedding transmission alone is not enough for privacy protection, so the authors combine embedding obfuscation, encryption, and secure key management for multimodal retrieval.

Research Gap (basic idea)

- **Evaluation:** Existing secure transmission schemes may provide confidentiality and authentication, but their runtime overhead on edge devices remains an important factor for real-time image or embedding transmission.
- Metrics : CPU usage, memory usage, end-to-end latency, payload size, throughput.
- Other Metrics: encryption time, decryption time, key setup/exchange time